Can Deep Learning Models Learn EEG Representations that Reflect Integrated Information and Distinguish Different Human Consciousness States?

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1 Introduction

One of the primary and most important challenges in neuroscience is understanding human consciousness. There have been multiple attempts to theorize consciousness over the past few decades. With the advent of Artificial Intelligence and neural networks that have been inspired by how the brain learns, we are closer to understanding the intricate mechanisms of the human brain.

Integrated Information Theory (IIT) is one such proposed theory that claims that consciousness is identical to a certain kind of information, the realization of which requires physical, not merely functional, integration, and which can be measured mathematically according to the phi metric [1]. According to this theory, different brain levels should exhibit varying levels of information integration.

Electroencephalography (EEG) is one such way to measure brain activity with high resolution. This project aims to bridge integrated information and deep learning, specifically neural networks, which can learn representations from complicated signals such as EEG. However, it is still an open question whether these learned representations align with IIT principles.

2 Objectives

The main objective of this project is to determine whether deep learning models can learn representations of EEG data that help distinguish between different human consciousness states (wake, REM, NREM), and reflect underlying IIT patterns.

Specifically, the project aims to:

1. Train a deep learning model to classify sleep stages from EEG data obtained from the Sleep-EDF dataset.
2. Compute primary IIT-related metrics such as mutual information and correlation.
3. Compare learned representations with these metrics to evaluate whether models capture integrated structure.

3 Methodology

3.1 Data Acquisition and Preprocessing

We will use the publicly available Sleep-EDF dataset [2]. The dataset contains multichannel EEG recordings across different sleep stages. A bandpass filter (0.5–30 Hz) using a zero-phase Butterworth filter will be applied to remove drift and high-frequency noise [3].

3.2 Feature Construction and Model Training

Raw EEG time-series segments will be used as inputs to deep learning models. These models will be trained to classify brain states. Model performance will be evaluated based on classification accuracy and other relevant metrics.

3.3 Learned Latent Representation Analysis

Latent representations from hidden layers will be extracted and visualized using dimensionality reduction techniques. This will help examine patterns associated with different consciousness states.

3.4 Information Theory Comparison

Information-theoretic measures such as mutual information and correlation will be computed for EEG segments. These will act as proxies for integrated information. Their relationship with learned representations will be analyzed to assess whether higher integration corresponds to more separable representations.

4 Expected Outcomes

We expect that:

- Deep learning models will successfully classify different brain states from EEG data.
- Higher levels of information integration (e.g., wakefulness) will correlate with distinct representation patterns, providing indirect support for IIT-inspired interpretations.

This study may serve as a proof-of-concept for future research exploring how AI models can contribute to computational studies of consciousness.

5 References

References

[1] https://iep.utm.edu/integrated-information-theory-of-consciousness/

[2] https://www.physionet.org/content/sleep-edfx/1.0.0/

[3] Dobbin, Emma. *Dual Architecture of Consciousness Transitions: Evidence from Sleep EEG Analysis*. Consciousness Gradient Theory Group Ltd, United Kingdom.